let's playte

The Social Plate Ranking Game

Product Requirements Document · v3.0

Version	3.0 — Final pre-engineering draft
Date	2025
Status	Ready for Engineering
Platform	Web App — Mobile-First (iOS + Android share)
Brand	playte (always lowercase)

1. Product Overview

Playte is a real-time multiplayer plate-ranking web app for groups dining out together. The host creates a session, searches for the restaurant, adds the dishes the table ordered, and shares a join code with the group. Every player drags dishes into their personal ranking order privately. Once all players have submitted, the app automatically advances to the Flavor Journey — a three-page results experience showing group dish rankings, a player-by-player comparison, and four branded insight cards. Accounts are optional; all data is stored for potential future analytics and recommendations.

1.1 Problem Statement

After a group meal, everyone has opinions about what was best — but those opinions evaporate before the check arrives. There is no easy way to capture, compare, and revisit collective and individual taste preferences in the moment.

1.2 Solution

A fast, no-install web app that turns post-meal dish debate into a structured, shareable game. The result is a personalized "Flavor Journey" — a snapshot of the group's taste that lives in account history and can be shared to social platforms.

1.3 Target Users

- Primary: Groups of 2–20 friends or family dining out together
- Secondary: Repeat diners who want a history of restaurant experiences and rankings

- Tertiary: Anyone interested in tracking and comparing taste preferences over time

2. Goals & Success Metrics

2.1 Business Goals

- Build a shareable dining experience that spreads organically via iOS/Android share
- Convert guest players to account holders through the end-of-game save prompt
- Accumulate structured restaurant + dish + ranking data as a foundation for future features (recommendations, analytics)

2.2 Success Metrics

Metric	Target	Timeframe
Sessions completed per week	500+	3 months post-launch
Average players per session	4+	Ongoing
Guest-to-account conversion	25%+	Ongoing
Session completion rate (all players submit)	75%+	Ongoing
Flavor Journey shares per session	1.5+ avg	Ongoing
Return sessions (same account, 2+ plays)	40%+	6 months

3. User Stories

3.1 Host Flow — Starting a Game

ID	As a...	I want to...	So that...	P
US-01	Host	Enter my display name	Players know who I am	P0
US-02	Host	Search for the restaurant by name and select from results	The session is linked to a real, stored restaurant record	P0
US-03	Host	Add dish names one by one to the session	Players know what they are ranking	P0
US-04	Host	Receive a join code to share with my group	Others can join without friction	P0
US-05	Host	See a count of how many players have joined (not their names)	I know when enough people are in	P0

US-06	Host	Start the ranking phase when ready	The game begins at the right time	P0

3.2 Player Flow — Joining a Game

ID	As a...	I want to...	So that...	P
US-07	Player	Enter my display name	My friends know who I am in the results	P0
US-08	Player	Enter a join code to join a session (no account required)	I can play immediately	P0
US-09	Player	See a waiting screen until the host starts ranking	I know I have joined successfully	P0
US-10	Player	Drag dishes to reorder them from best (#1) to least liked	My ranking is recorded correctly	P0
US-11	Player	Submit my full ranking privately	No one sees my order before the reveal	P0
US-12	Player	See the loading screen while waiting for all others to submit	I know the game is processing	P0

3.3 Results — Flavor Journey (3 Pages)

The Flavor Journey auto-advances once all players have submitted rankings. No host action required. Players can also navigate back through the three pages. Each page has a Share button (native OS share sheet) and, if the viewer is a guest, a 'Save to Account' button.

Page 1 — Group Dish Rankings

ID	Story	P
US-13	See an animated podium of stacked plates (gold/silver/bronze) for the top 3 ranked dishes	P0
US-14	See the full numbered list of all dishes ranked #1 to #N by group consensus (average rank position)	P0
US-15	Tap 'Start Flavor Journey' to navigate to the Playters page	P0

US-16	Tap Share to open the native OS share sheet for this results card	P1

Page 2 — Playters (Player Comparison)

ID	Story	P
US-17	See a list of all players with display name, avatar/icon, and % match to my own rankings	P0
US-18	Tap any player card to expand and see the full ranked dish list they submitted	P0
US-19	Tap 'Share Flavor Journey' or 'See Flavor Journey' to proceed to the Insights page	P0

Page 3 — Flavor Journey (Insights)

ID	Story	P
US-20	See 'Most Loved' 🏆: the dish with the lowest average rank position across all players (group's favorite)	P0
US-21	See 'Nacho Type' 🤔: the dish the group liked least — highest average rank position (least preferred overall)	P0
US-22	See 'Hot & Cold' 😤: the most controversial dish — highest variance in rank positions across all players	P0
US-23	See 'Best (Taste) Buds': the other player whose full ranking matches mine most closely (per-player, Spearman correlation)	P0
US-24	Tap Share to open the native OS share sheet for the Flavor Journey card	P1
US-25	If I am a guest: see a 'Save to Account' button that triggers account creation / login, then attaches this session	P1
US-26	If I am logged in: session is automatically saved; no manual save button shown	P1

3.4 Restaurant Search

ID	As a...	I want to...	So that...	P

US-27	Host	Type a restaurant name and see matching suggestions (via Google Places API or similar)	I select the correct restaurant record rather than free-typing	P0
US-28	Host	See restaurant name, address, and optionally cuisine type in the suggestion list	I can confirm I have the right place	P0
US-29	Host	Select a restaurant from results to attach it to the session	The session is linked to structured restaurant data (name, address, place ID)	P0
US-30	Host	Fall back to manual entry if the restaurant is not found in search	No session is blocked by a missing search result	P1

3.5 Accounts & History

ID	As a...	I want to...	So that...	P
US-31	Any user	Create an account with email/ password or Google OAuth	I can save session history	P1
US-32	Logged-in user	Have sessions I play automatically attached to my account	History builds without extra effort	P1
US-33	Account holder	Browse past sessions by date and restaurant	I can relive dining memories	P1
US-34	Guest player	Tap 'Save to Account' at the end to claim the session	I don't lose the memory even if I played as a guest	P1

4. Functional Requirements

4.1 Session Lifecycle

Sessions move through four states. The transition from RANKING → RESULTS is automatic (no host action), triggered when all players have submitted.

State	Trigger	Who can act	Notes
LOBBY	Host creates session	Host only	Host adds dishes; players join; host sees join count

RANKING	Host taps 'Start' in lobby	All players	Dish list locked; each player submits privately
RESULTS	All players submit — auto-advance	All (read-only)	Flavor Journey revealed; no further input
ARCHIVED	24h expiry or manual — future admin tool	N/A	Session stored permanently for data retrieval

Host disconnect policy: if the host leaves before ranking starts, session data may be lost — this is acceptable. Once status = RANKING, all state persists server-side regardless of host connectivity.

4.2 Host Setup Flow (Screen by Screen)

- Screen 1 — 'what's your name?': text input, 'next' CTA
- Screen 2 — 'where are you dining?': restaurant search input → suggestion list → select → 'next'; fallback to manual entry
- Screen 3 — 'what did you order?': enter dishes one at a time; they appear in a list below; 'next' CTA
- Screen 4 — Join Code: large join code displayed; player count shown (e.g. '3 players joined'); 'let's playte' starts ranking

4.3 Player Join Flow (Screen by Screen)

- Screen 1 — 'what's your name?': text input, 'next' CTA
- Screen 2 — 'join a game': code input (numeric keypad on mobile), 'join' CTA
- Screen 3 — Waiting: 'your table is almost ready...' — held until host triggers ranking

4.4 Ranking Interface

- Header: 'rank your dishes' / subtext: 'drag to order, favorite on the top!'
- All dishes shown as draggable cards; drag handle (×) on each row
- Full ordering required — every dish must be placed; no partial rankings
- Rankings remain private until all players have submitted
- Once submitted, rankings are locked — no edits
- After submitting, player sees the red full-screen 'mixing up your flavors...' loading screen with bowl animation
- Page auto-refreshes / advances to Flavor Journey when the last player submits

4.5 Join Code

The default is a short numeric code matching the Figma design. Engineering may propose an alternative (e.g. a short alphanumeric code, a 6-character word pair, or a QR code alongside the numeric code) if there is a more robust or collision-resistant approach. The constraint is that it must be easy to share verbally at a table.

- Code must be unique among all active (non-archived) sessions
- Codes expire 24 hours after session creation or when session is archived
- Numeric keypad UI on mobile for code entry

4.6 Restaurant Search & Data Storage

- Use Google Places API (or equivalent) for restaurant search autocomplete
- Store the selected restaurant as a canonical record in the restaurants table: name, address, city, state, country, lat/lng, external_place_id
- All sessions, dishes, and rankings are linked to this restaurant record
- This data model enables future retrieval of: all dishes ranked at a restaurant, average rankings per dish across all sessions, most popular dishes by restaurant or cuisine type
- If the host falls back to manual entry, a restaurant record is still created — it just lacks the external place ID and coordinates

4.7 Scoring & Insight Algorithms

Group Dish Rankings (Page 1): Dishes sorted by average rank_position across all players. Lower average = higher rank. Tiebreaker: count of #1 votes.

Most Loved: Dish with the lowest average rank position (most consistently ranked near the top). Tiebreaker: lowest variance.

Nacho Type: The group's least liked dish — highest average rank position (most consistently ranked toward the bottom).

Hot & Cold: The most controversial dish — highest variance in rank positions across all players. A dish where some loved it and some hated it.

Best (Taste) Buds (per-player): The other player whose full ranking order has the highest Spearman rank correlation with the current player's ranking. Each player sees their own unique Best Bud.

% Match (Playters page, per-player): Derived from the Spearman correlation between two players' rankings, normalized to a 0–100% scale.

4.8 Share Feature

- Each of the 3 Flavor Journey pages has a Share button
- Uses the Web Share API (navigator.share) on iOS Safari and Android Chrome
- Share payload: a pre-composed shareable image/card of the results + short caption + session link
- Desktop fallback: copy link to clipboard

4.9 Save to Account

- 'Save to Account' button is only shown to guest players
- For logged-in users: session is automatically saved to their account — no button shown
- Tapping 'Save to Account' triggers the account creation / login flow, then retroactively attaches the session to the new account

4.10 Authentication

- Supported methods: email + password, Google OAuth
- Account creation is always optional — full game playable as guest
- JWT-based auth tokens with refresh token rotation

4.11 Lobby Player Count

- The host sees a count of players who have joined (e.g. '4 players joined') — not their names
- Players do not see each other in the lobby
- Player names are revealed for the first time on the Playters results page

4.12 Session & Game Limits

- Maximum 20 players per session
- Maximum 20 dishes per session

5. Screen Inventory

Screen	Route	Auth	Description
Landing	/	No	'let's playte' — 'start a game' and 'join a game' CTAs
Host — Name	/create/name	No	'what's your name?' — text input
Host — Restaurant	/create/restaurant	No	Restaurant search input with autocomplete suggestions; manual fallback

Host — Dishes	/create/dishes	No	'what did you order?' — add items one at a time
Host — Lobby	/session/:id/lobby	No	Large join code + player count; 'let's playte' starts ranking
Player — Name	/join/name	No	'what's your name?' — text input
Player — Code Entry	/join/code	No	'join a game' — numeric code entry
Waiting Room	/session/:id/waiting	No	'your table is almost ready...' — held until host starts ranking
Ranking	/session/:id/rank	No	Drag-to-rank all dishes; submit button; private
Loading	/session/:id/loading	No	Red 'mixing up your flavors...' animation; auto-advances when all submit
Results — Dishes	/session/:id/results/dishes	No	Group dish podium + full ranked list; 'Start Flavor Journey'
Results — Playters	/session/:id/results/players	No	Player list with % match; tap to see their ranking
Results — Insights	/session/:id/results/insights	No	Most Loved / Nacho Type / Hot & Cold / Best Buds; Share + Save (guest only)
Auth	/auth	No	Email + Google OAuth sign-up / login
Profile / History	/profile	Yes	Past sessions list by date and restaurant
Session Detail	/session/:id	Yes	Full Flavor Journey replay for a past session
404 / Error	*	No	Friendly error state

6. Data Storage Strategy

All session data is stored in a structured format to support future retrieval and analytics. The schema is designed around canonical restaurant records so that dish rankings can be aggregated across sessions.

6.1 What Is Stored

- Restaurants: name, address, city, state, country, lat/lng, Google Place ID (or equivalent), cuisine type
- Dishes: name, linked to both a session and a restaurant record
- Sessions: restaurant, host, status, all timestamps
- Players: display name, user account link (if any), guest token (if guest)
- Rankings: every player's rank_position for every dish, with submission timestamp
- Insights: computed results cached per session (avg ranks, variance, correlations, per-player insights)

6.2 Future Retrieval Use Cases

- All sessions played at Restaurant X
- Average group ranking for Dish Y across all sessions at Restaurant X
- Most popular dishes across all playte sessions globally
- A user's personal taste profile: dishes they consistently rank highly
- Restaurant or dish recommendations based on past rankings

7. Non-Functional Requirements

7.1 Performance

- Ranking page auto-advance to Flavor Journey must occur within 3 seconds of the last player submitting
- Restaurant search autocomplete must return results within 500ms
- Insights computation must complete within 3 seconds of session close
- App must be usable on restaurant cellular (3G minimum target)

7.2 Compatibility

- Primary: iOS Safari, Android Chrome (mobile-first)
- Secondary: Desktop Chrome, Firefox, Safari, Edge (latest 2 versions)
- Share: Web Share API on iOS + Android; copy-link fallback on desktop
- No native app install required

7.3 Accessibility

- WCAG 2.1 AA compliance
- Drag-to-rank must have a keyboard-accessible fallback (arrow keys or numbered tap-to-move)

7.4 Security

- All traffic over HTTPS
- Passwords hashed with bcrypt (min cost factor 12)
- Rate limiting on session creation and join endpoints
- Join codes expire after 24 hours or on archive

7.5 Data Retention

- Guest sessions: retained for 90 days, then purged
- Authenticated sessions: retained indefinitely while account is active
- Account deletion: soft-delete with async anonymization of related data (GDPR)

8. Future Considerations (Out of Scope v1)

Feature	Notes
Dish photos / image upload	Adds storage complexity; not in Figma
Animated share image card	High social value; needs canvas/image generation
Push / email notifications	Useful for reminders; not MVP-critical
Cross-session taste analytics	Requires longitudinal data accumulation first
Friend graph / leaderboards	Requires social layer
Restaurant recommendations	Requires ML; separate product track
Cuisine / dish tagging	Would enrich future analytics
Admin / moderation tools	Needed at scale

9. Open Questions

#	Question	Owner	Status
Q1	Join code format: engineering to propose the most collision-resistant short code approach (numeric, alphanumeric, word pair, etc.) that is still easy to share verbally at a table.	Engineering	Open
Q2	Restaurant search API: confirm Google Places vs. alternative (Yelp Fusion, Foursquare). Consider cost, rate limits, and terms.	Engineering	Open
Q3	Share image card: is the shareable image a server-rendered OG image (static per session), a client-side canvas screenshot, or a plain text + link share? This affects engineering complexity.	Engineering / Design	Open
Q4	What happens if a player closes their browser mid-ranking before submitting? Does this block the auto-advance? Propose: a timeout (e.g. 10 min) after which a non-submitted player is dropped.	Product	Open
Q5	Lobby: should the host be able to kick a player who joined by mistake?	Product	Open

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